Evaluating Deployed-Selection Robustness in Forecasting Benchmarks

Anonymous Authors[1]

Abstract

Forecasting benchmarks commonly rank models by forecast-side metrics, and these metrics remain the appropriate object when the target is predictive quality. In deployment-facing forecasting settings, however, forecasts may be consumed through a fixed forecast-to-decision interface, such as a threshold alert, hysteresis threshold, or top-k budget alert, before deployed-side scoring. We do not propose a new forecaster, training objective, or universal deployed metric; instead, we study whether forecast-side ranking preserves deployed-side model selection under the same fixed interface. Across a synthetic zero-friction anchor, forecasting-native event-micro and Traffic-Hourly tasks, and inventory corroboration, we find that forecast-side winners can become recurrently deployed-suboptimal in moderate-to-high friction regimes in the studied settings. The implication is not to invalidate forecast-side evaluation for prediction, but to report fixed-interface deployed-selection robustness alongside forecast-side metrics when forecasting benchmarks are used for deployment-facing model selection.

1. Introduction

Forecasting benchmarks typically compare models using forecast-side metrics, and those metrics are the right object when the benchmark asks about predictive quality. Some deployment-facing forecasting evaluations, however, consume forecasts through a fixed forecast-to-decision interface before deployed-side scoring. By a fixed interface we mean a common forecast-to-action rule applied to every forecaster before scoring, such as triggering an alert when $p_t > \tau$, changing state only if the score moves by at least δ, or issuing the top-k alerts under a fixed budget. When that interface includes switching or deployment friction, forecast-side ranking can diverge from deployed-side model selection: the model that wins on the forecast metric need not remain the model that performs best after the fixed interface is applied. This is not an argument for replacing forecast benchmarks with a single deployed metric. Because deployed metrics depend on a specific interface, simulator, and friction model, they are not a drop-in replacement for general forecasting benchmarks. Our claim is narrower: deployment-facing benchmarks should report whether forecast-side ranking preserves deployed-side model selection under the chosen fixed interface.

This paper frames the mismatch as a reporting diagnostic for deployment-facing forecasting benchmarks. We call the diagnostic fixed-interface deployed-selection robustness: whether the forecast-side winner remains the deployed-side winner after forecasts are passed through the same fixed interface. Our main empirical question, Q2, asks whether this preservation holds when the interface is fixed and the forecaster varies. Q1 is retained only as mechanism support: it holds the proposed action path fixed and asks how frictional execution can separate proposed and realized actions. This framing keeps forecast-side metrics as prediction measures while asking what deployment-facing evaluation should report alongside them.

We frame this diagnostic as a lightweight report-card protocol for deployment-facing forecasting benchmarks: given forecasts, a fixed forecast-to-decision interface, and a friction grid, the protocol reports forecast-side winners, deployed-side winners, agreement rates, deployed-suboptimal shares, paired deployed-utility gaps, and tie/uncertainty diagnostics.

This paper makes three focused contributions.

- First, we formulate fixed-interface deployed-selection robustness as a lightweight report-card protocol for deployment-facing forecasting benchmarks: under a specified fixed interface and friction grid, the report card records whether forecast-side model selection is preserved after deployment-side scoring.
- Second, across the studied fixed interfaces, we show that forecast-side winners can become recurrently deployed-suboptimal in moderate-to-high friction regimes.
- Third, we organize evidence in a Q2-first structure, with event-micro and Traffic-Hourly as main forecasting-native

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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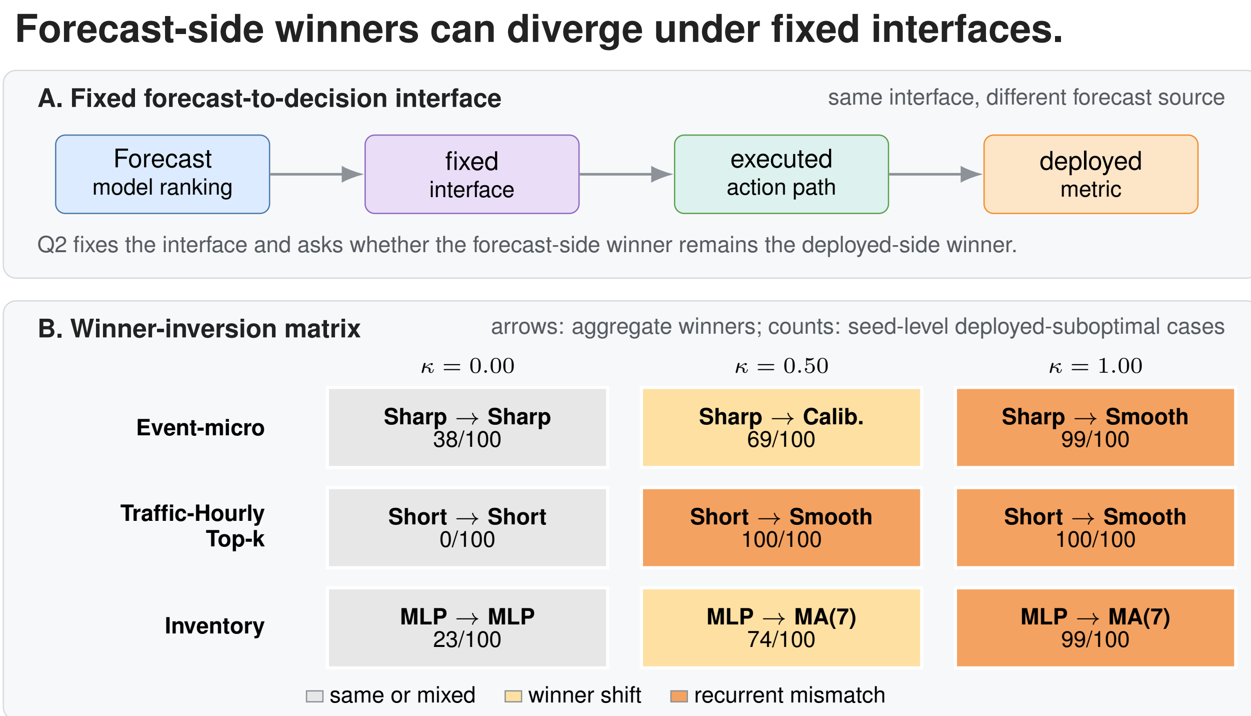

Figure 1. Fixed-interface evaluation can change model selection. Panel A shows the Q2 setting: the interface is fixed, the forecaster varies, and the question is whether the forecast-side winner remains the deployed-side winner. Panel B reports aggregate forecast-side and deployed-side winners across domains and friction levels. Cell counts denote seed-level deployed-suboptimal cases for the forecast-selected model; darker orange marks recurrent mismatch. Gray zero-friction cells are included for completeness rather than interpreted as friction-causal anchors; the clean zero-friction anchors are the synthetic sanity row in Table 1 and the Traffic-Hourly Top-k zero-friction cell.

evidence, inventory as operational corroboration, and Q1 only as mechanism support.

We use Q1/Q2 only to separate mechanism from model-selection evidence. Q1 fixes the proposed action path and varies the interface as an exact-control check. Q2 fixes the interface and varies the forecaster as the main model-selection question.

2. Results

2.1. Q2: Fixed interface and deployed-selection robustness

Q2 asks the paper's headline reporting question: under one fixed forecast-to-decision interface, does the forecast-side winner remain the deployed-side winner? Table 1 reports the resulting deployed-selection robustness report card. The table is not merely a performance summary; it is the output format of the proposed report-card diagnostic. The zero-friction synthetic row is a sanity anchor. Event-micro and Traffic-Hourly provide forecasting-native Q2 evidence under two different fixed-interface structures, while inventory is included only as operational corroboration. Figure 1 visualizes the main winner-inversion evidence across the forecasting-native and corroboration rows.

In the report card, Event-micro shows the clearest primary Q2 pattern: the forecast-side winner remains Reactive sharp, but the deployed winner changes to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The deployed-selection consequence is recurrent: the forecast-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The event-micro result is not confined to the canonical threshold or Brier ranking: at frictions 0.50/1.00, deployed-suboptimality remains 77/100 and 98/100 under an alternate threshold, 74/100 and 99/100 under log-loss reranking, and 55/100 and 82/100 under a hysteresis interface, where the direction is preserved with smaller magnitude.

A second forecasting-native Traffic-Hourly Top-k alert task repeats the same selection failure under a fixed-budget alert interface. In the selected Top-k setting ($k = 249$), Reactive short is both the forecast-side winner and the deployed winner at zero friction, with exact agreement 1.0. Once switching costs are introduced, however, the forecast-side winner remains Reactive short while the deployed winner shifts to Lagged smoother, and deployed-suboptimality reaches

Table 1. Deployed-selection robustness report card, the output format of the proposed diagnostic. Each row reports one domain $\times$ fixed interface $\times$ friction cell and asks whether the forecast-side winner remains the deployed-side winner after the same interface is applied. Subopt. reports the seed-level deployed-suboptimal share/count; the 95% CI reports the paired-bootstrap interval for the mean deployed gap.

Domain	Interface	κ	Forecast winner	Deployed winner	Agree.	Subopt.	Mean gap	95% CI	Role
Synthetic	zero-friction anchor	0.00	Naive last	Naive last	1.00	0.00 (0/20)	0.000	[0.000, 0.000]	Sanity
Event-micro	threshold τ=0.55	0.50	Reactive sharp	Calibrated	0.31	0.69 (69/100)	0.011	[0.009, 0.014]	Primary
Event-micro	threshold τ=0.55	1.00	Reactive sharp	Smoother	0.01	0.99 (99/100)	0.057	[0.052, 0.063]	Primary
Traffic Top-k	budget k=249	0.50	Reactive short	Smoother	0.00	1.00 (100/100)	7.016	[6.756, 7.283]	Second
Traffic Top-k	budget k=249	1.00	Reactive short	Smoother	0.00	1.00 (100/100)	24.651	[24.226, 25.079]	Second
Inventory	replenishment	1.00	Small MLP	MA(7)	0.01	0.99 (99/100)	1.029	[0.946, 1.111]	Corrob.

$100/100$ seeds at both friction 0.5 and friction 1.0. Additional Traffic checks support this reading most strongly for the Top-k interface family: across $k \in \{125, 249, 500\}$, zero-friction rows remain aligned and positive-friction rows keep the reactive forecast-side winner deployed-suboptimal; predefined time-of-day and relative-rank variants are kept in the appendix rather than used to broaden the main claim.

An ϵ-tie audit gives the same qualitative reading for the main forecasting-native rows: Traffic-Hourly Top-k and Event-micro at $\kappa \in \{0.50, 1.00\}$ remain deployed-suboptimal under relative tie thresholds up to $\epsilon_{\mathrm{rel}} = 0.005$; inventory remains corroborative and lower-margin secondary checks are reported only in the appendix.

Inventory is used only as operational corroboration, not as main forecasting-native evidence. At friction 0.50 and 1.00, the forecast-selected Small MLP is deployed-suboptimal in 74/100 and 99/100 seeds, respectively, under the fixed replenishment interface; the report card includes only the high-friction inventory row to keep this role bounded.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

Appendix robustness. Appendix checks surface robustness without changing the evidence hierarchy: event-micro includes alternate-threshold, log-loss, and hysteresis-interface checks; Traffic-Hourly includes Top-k interface-family, predefined time-of-day slice, and relative-rank checks; and expanded inventory and load-following remain fairness or appendix-only context. These checks are consistent with the main qualitative conclusion: under a fixed interface, a forecast-side winner can remain forecast-best while becoming recurrently deployed-suboptimal in moderate-to-high friction regimes.

Why forecast ordering need not be preserved. Proper forecast-side scores evaluate predictive quality, not the composed object obtained after forecasts are mapped to actions and executed through a frictional interface. If a forecaster's forecast-side advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. This is why Q1 is useful as mechanism support and why Q2 is the actual benchmark-selection question.

3. Discussion, Related Work, and Limitations

Forecasting benchmarks and evaluation frameworks. Forecasting benchmarks and evaluation frameworks now span live AI forecasting, broad time-series archives, foundation-model evaluations, text-conditioned forecasting, and standardized TSFM evaluation pipelines (Karger et al., 2025; Yang et al., 2025; Aksu et al., 2024; Godahewa et al., 2021; Das et al., 2024; Williams et al., 2025; Li et al., 2025; Goktas et al., 2026; Zhao et al., 2025). These works compare forecasting capability, datasets, model families, or evaluation pipelines. Our question is complementary: when a benchmark is interpreted as deployment-facing model-selection advice under a fixed forecast-to-decision interface, should it also report whether the forecast-side winner remains the deployed-side winner?

Benchmark variance and uncertainty reporting. Our reporting emphasis is closer to work on benchmark variance, aggregate uncertainty, and leaderboard interpretation. Prior work argues that benchmark conclusions should account for variance across training and evaluation choices (Bouthillier et al., 2021), quantify uncertainty in aggregate benchmark metrics (Longjohn et al., 2025), and use principled aggregation with bootstrap intervals in realistic forecasting benchmarks (Shchur et al., 2025). Recent re-evaluations of long-term time-series forecasting further caution against over-reading leaderboard differences as stable model dominance (Brigato et al., 2026). We adopt the same spirit, but the object of reporting is different: whether forecast-side selection is preserved after a fixed forecast-to-decision interface.

Forecast scoring and decision-focused learning. This is distinct from proper scoring rules, decision-maker-perspective forecast evaluation, decision-focused learning, and predict-then-optimize, which either define forecast quality or modify learning or optimization objectives for downstream decisions (Gneiting and Raftery, 2007; Murphy, 1993; Raeth and Ludwig, 2025; Donti et al., 2017; Elmachtoub and Grigas, 2022; Mandi et al., 2024). Here, forecasters and the interface are fixed; the contribution is a reporting diagnostic for whether forecast-side model selection survives the composed forecast-to-interface-to-utility evaluation.

Discussion and limitations. Our claim is evaluative and deliberately narrow. We do not claim that forecast-side metrics are invalid for prediction, and we do not seek a single reporting rule for all forecasting applications. We use low, moderate, and high only as shorthand for the lower, middle, and upper parts of each domain's tested friction grid; these labels are not intended as a universal calibration across domains. Event-micro provides the main forecasting-native evidence, Traffic-Hourly provides a second forecasting-native task family, and inventory provides operational corroboration, but this is still not a comprehensive benchmark collection. Broader inventory sweeps are reported in the appendix, and load-following is secondary; the strongest main-text evidence comes from the moderate-to-high friction settings studied here. The intended artifact is modest: a report-card template for deployment-facing model selection under a specified fixed interface, not a new benchmark suite or universal deployed metric. Under this interpretation, inventory and load-following remain corroborative or secondary context rather than standalone statistical evidence for a general effect.

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Reproducible report-card checklist. For each task family, the report-card artifact records:

1. task definition, seed count, and evaluation split;
2. forecast-side metric and tie-breaking rule;
3. fixed forecast-to-decision interface and interface parameters;
4. friction grid, including a zero-friction anchor and its status when applicable;
5. candidate model families and representative selection rule;
6. split/seed manifest;
7. forecast-side winner and deployed-side winner for each domain $\times$ interface $\times$ friction cell;
8. agreement rate and deployed-suboptimal share/count of the forecast-selected model;
9. mean and median paired deployed-utility gaps with uncertainty intervals;
10. uncertainty and ϵ-tie audit;
11. mixed or failed-gate variants, when applicable.

Mixed and failed-gate variants. We retain mixed and failed-gate variants as audit evidence rather than promoting them into the main report card. This reduces the risk that the report card is read as a collection of selected successes: variants that fail the zero-friction alignment check, remain lower-margin under tie diagnostics, or show only partial recurrence are kept as appendix context or excluded from main-text claims.

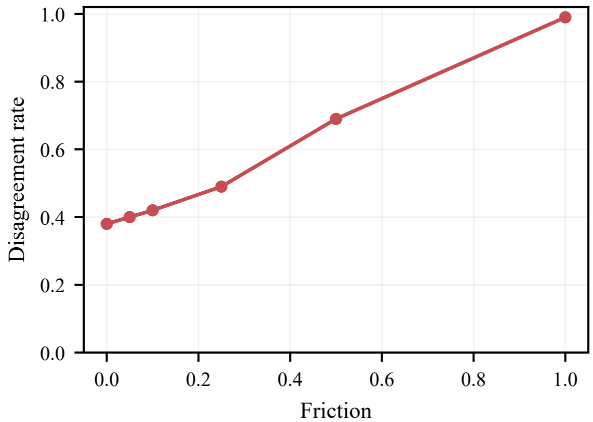
Table A1. Mixed and failed-gate variants retained as audit context. These variants are not promoted into the main report card because they either fail the clean zero-friction alignment check, show lower-margin or tie-sensitive behavior, or provide only partial recurrence.

Variant	Gate outcome	Interpretation	Placement
Traffic daytime $\kappa = 0.5$	Partial	Lower-margin, directionally consistent	Appendix only
Inventory zero/low friction	Mixed	Not a main zero-friction anchor	Appendix only
Expanded inventory	Mixed	Fairness context only	Appendix only
Load-following	Tie-sensitive	Directional secondary check	Appendix only

These variants are useful because they separate friction-induced deployed-selection drift from other sources of mismatch, such as objective mismatch, weak margins, or insufficient seed support. We therefore use them to bound the interpretation rather than to strengthen the main claim.

A. Event-Micro Robustness Package

Because event-micro provides the paper's main forecasting-native Q2 evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal forecasting-native binary-event setting evaluated over 100 seeds, in which each forecaster emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical forecast-side criterion. This appendix records the canonical six-row Brier summary plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.



κ	Forecast	Deployed	Agree.	Gap
0.00	Sharp	Sharp	0.62	0.003
0.05	Sharp	Sharp	0.60	0.002
0.10	Sharp	Sharp	0.58	0.002
0.25	Sharp	Calibrated	0.51	0.003
0.50	Sharp	Calibrated	0.31	0.011
1.00	Sharp	Smoother	0.01	0.057

Figure A1. Event-micro canonical summary. Left: disagreement between forecast-side and deployed-side model selection increases with friction. Right: canonical fixed-threshold selection summary; gap is the mean paired deployed shortfall of the forecast-selected model.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A2. Alternate-threshold robustness for the event-micro benchmark. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A3. Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative winner drift and recurrent deployed-suboptimality pattern persists when forecast-side ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A4. Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative winner drift and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative winner-drift and recurrent deployed-suboptimality pattern persists in a minimal forecasting-native event-micro benchmark evaluated over 100 seeds.

A.1. Interface-Aware Event-Micro Stats Addendum

This addendum records interface-specific recurrence and gap summaries for the hysteresis-threshold check without changing the main cross-domain recurrence package.

Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Fixed threshold	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Fixed threshold	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Hysteresis threshold	0.50	100	0.45	0.55 (55/100)	[0.452, 0.644]	Wilson
Hysteresis threshold	1.00	100	0.18	0.82 (82/100)	[0.733, 0.883]	Wilson

Table A5. Interface-aware recurrence and share uncertainty for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

The aggregate recurrence table reports the seed-level pattern directly, replacing the dense seed-strip visualization.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A6. Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

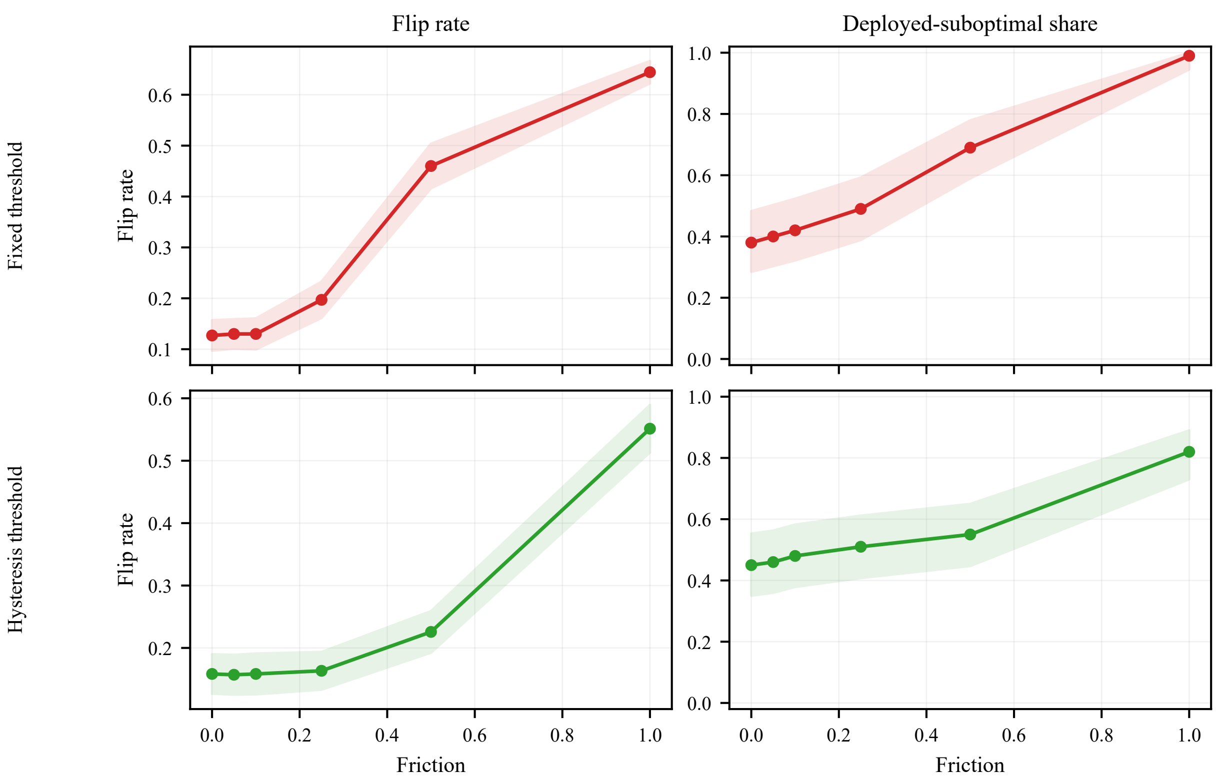


Figure A2. Event-micro robustness across fixed-threshold and hysteresis interfaces. Rows denote interfaces and columns denote disagreement and deployed-suboptimality metrics; hysteresis preserves the direction with attenuated magnitude.

B. Traffic-Hourly Forecasting-Native Checks

B.1. Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A7. Traffic-Hourly Top-k forecasting-native Q2 check under a fixed-budget interface. Agreement is strongest at zero friction and weakens at higher friction, where the deployed winner shifts away from the forecast-side winner.

In the selected Top-k setting, Reactive short is the forecast-side winner and the deployed winner at zero friction. At frictions 0.5 and 1.0, the deployed winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win on the forecast side. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive forecast-optimal family need not remain deployment-optimal once switching is penalized.

B.2. Top-k Interface-Family Sweep

k	κ	Forecast winner	Deployed winner	Subopt.	Mean gap	95% CI	Median gap	Verdict
125	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
125	0.50	Reactive short	Calibrated baseline	100/100	3.794	[3.584, 4.012]	3.725	Pass
125	1.00	Reactive short	Calibrated baseline	100/100	15.986	[15.639, 16.339]	15.955	Pass
249	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
249	0.50	Reactive short	Lagged smoother	100/100	7.016	[6.756, 7.283]	6.977	Pass
249	1.00	Reactive short	Lagged smoother	100/100	24.651	[24.226, 25.079]	24.473	Pass
500	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
500	0.50	Reactive short	Lagged smoother	100/100	14.314	[14.048, 14.574]	14.383	Pass
500	1.00	Reactive short	Lagged smoother	100/100	37.985	[37.521, 38.438]	38.080	Pass

Table A8. Traffic-Hourly Top-k interface-family sweep. The canonical $k = 249$ rows are reproduced from the recovered time-level Traffic source, and non-canonical $k = 125$ and $k = 500$ rows show the same qualitative pattern: zero-friction alignment and recurrent positive-friction deployed-suboptimality of the reactive forecast-side winner. Confidence intervals are paired-bootstrap intervals for the mean deployed gap.

Across $k \in \{125, 249, 500\}$, Reactive short remains both the forecast-side and deployed-side winner at zero friction. At positive friction, Reactive short remains forecast-best but becomes deployed-suboptimal in all matched seeds, with positive paired deployed gaps and confidence intervals bounded away from zero.

B.3. Predefined Time-of-Day Slice Check

Slices are fixed before outcome computation as overnight, daytime, and evening blocks. Switching costs are computed on the full 168-hour timeline and utilities are then aggregated over the slice time points, so the interface state is not reset within each slice.

Slice	κ	Forecast winner	Deployed winner	Subopt.	Mean gap	95% CI	Median gap	Verdict
Overnight	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Overnight	0.50	Reactive short	Lagged smoother	100/100	13.981	[13.681, 14.279]	14.087	Pass
Overnight	1.00	Reactive short	Lagged smoother	100/100	29.858	[29.289, 30.430]	30.141	Pass
Daytime	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Daytime	0.50	Reactive short	Calibrated baseline	66/100	1.731	[1.385, 2.085]	1.205	Partial
Daytime	1.00	Reactive short	Calibrated baseline	100/100	16.049	[15.323, 16.762]	16.087	Pass
Evening	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Evening	0.50	Reactive short	Calibrated baseline	100/100	10.086	[9.717, 10.444]	10.027	Pass
Evening	1.00	Reactive short	Calibrated baseline	100/100	33.269	[32.649, 33.873]	33.288	Pass

Table A9. Predefined Traffic-Hourly time-of-day slice check. Slices are fixed before outcome computation. Overnight and evening reproduce the positive-friction mismatch pattern at both positive-friction levels; daytime is directionally consistent but lower-margin at $\kappa = 0.5$. We therefore treat this slice check as appendix-only support. Confidence intervals are paired-bootstrap intervals for the mean deployed gap.

The slice check is supportive but weaker than the interface-family sweep. Overnight and evening pass at both positive-friction levels, while daytime passes at $\kappa = 1.0$ and remains directionally positive at $\kappa = 0.5$ but falls below the pre-specified deployed-suboptimality threshold. We therefore use this result only as appendix context rather than as a stronger main-text claim.

B.4. Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A10. Traffic-Hourly relative-rank appendix variant. The same qualitative separation between forecast-side and deployed-side selection appears under fixed-budget set action.

As an additional Traffic appendix variant, the relative-rank target shows the same qualitative separation under a different forecasting-native label construction. Reactive short remains the forecast-side winner, while Calibrated baseline becomes the deployed winner once switching costs are introduced, so we report this variant as an appendix forecasting-native check. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

C. Inventory Operational Corroboration and Fairness Context

C.1. Five-Family Inventory Corroboration

This appendix records the full five-family inventory summary that underlies the bounded main-text inventory evidence. The mixed zero and low-friction rows are reported here, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.77	0.025	23/100
0.25	Small MLP	Small MLP	0.76	0.039	24/100
0.50	Small MLP	Moving average (7)	0.26	0.240	74/100
1.00	Small MLP	Moving average (7)	0.01	1.029	99/100

Table A11. Inventory corroboration under a fixed replenishment interface. The zero/low-friction regime is mixed, while moderate-to-high friction shows recurrent deployed-side misselection.

C.2. Expanded-Family Inventory Fairness Context

The stronger-baseline sweep is included as a fairness check for the inventory corroboration. It contextualizes the inventory corroboration; the main claim is supported by the fixed-interface selection comparison in the main text.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	100	0.105	0.870	1.0000	0.467	Moving average (7) / Naive persistence	1.000	does not clear stricter gate

Table A12. Expanded same-interface inventory sweep. The table records broader-family status checks for inventory robustness.

Detailed tuning grids and representative selection are summarized compactly; the robustness table reports the resulting fixed-interface comparison.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A13. Expanded-family inventory Q2 robustness summary for the broader comparison set.

The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

D. Load-Following Appendix-Only Secondary Check

Load-following is a secondary operational check. It tests whether forecast-side and deployed-side selection can diverge outside the event-micro and Traffic-Hourly settings. Its evaluation units are disjoint client groups rather than independent random seeds.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal groups / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A14. Load-following secondary check over disjoint client-group evaluation units. Near-zero friction remains mixed, while positive-friction regimes can favor a different deployed winner.

The result is consistent with the narrower reading used throughout the paper: deployed-side selection can differ from forecast-side selection under friction, but this domain is not used as primary evidence.

E. Q1 Mechanism Support

Q1 serves only as mechanism support for why Q2 divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

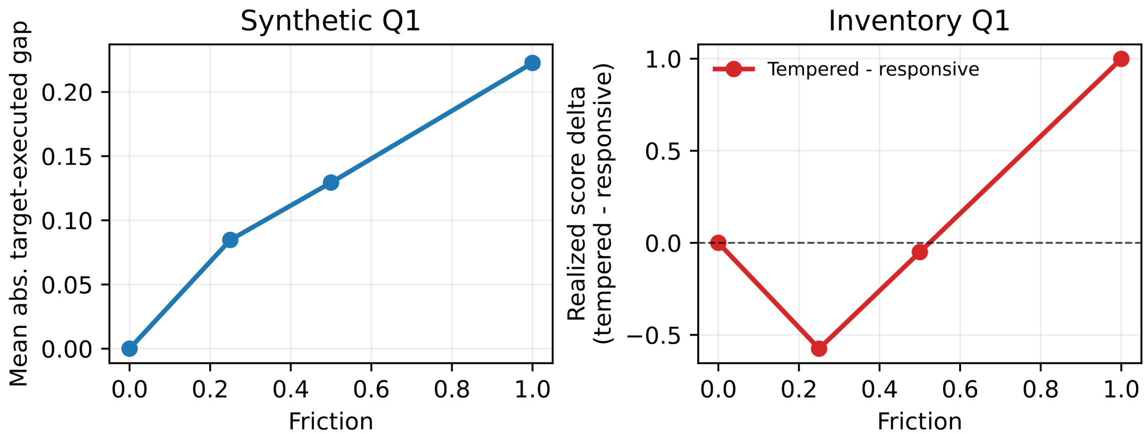


Figure A3. Q1 mechanism support. Even with the same proposed actions, a fixed frictional interface can alter realized actions and realized score; this explains how Q2 ranking divergence can arise.

F. Uncertainty Reporting for Seed-Level Recurrence and Deployed Gaps

This appendix reports uncertainty for the seed-level recurrence and paired deployed-gap summaries used in the main report card. We emphasize intervals and paired shortfalls rather than hypothesis-test significance. For binary share summaries, rows with fewer than 30 matched seeds use Clopper–Pearson exact intervals; rows with at least 30 matched seeds use Wilson intervals. For continuous deployed gaps, we use 10,000 paired seed bootstrap resamples. In the Load-following rows, the seed field denotes disjoint client-group evaluation units. We report these intervals as descriptive uncertainty summaries for the reporting diagnostic, not as hypothesis-test evidence for a general effect.

Domain	Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Event-micro	threshold $\tau = 0.55$	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Event-micro	threshold $\tau = 0.55$	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Traffic Top-k	budget $k = 249$	0.50	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Traffic Top-k	budget $k = 249$	1.00	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Inventory	replenishment	0.50	100	0.26	0.74 (74/100)	[0.646, 0.816]	Wilson
Inventory	replenishment	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Load-following	secondary	0.25	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper–Pearson
Load-following	secondary	0.50	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper–Pearson
Load-following	secondary	1.00	10	0.20	0.80 (8/10)	[0.444, 0.975]	Clopper–Pearson

Table A15. Seed-level recurrence and share uncertainty for deployed-suboptimality. Agreement rate is the fraction of matched seeds for which the forecast-side selected model is also the deployed-side best model; deployed-suboptimal share is the complementary fraction. We report Clopper–Pearson exact intervals for rows with $n < 30$ and Wilson intervals otherwise.

Aggregate recurrence intervals and uncertainty bands summarize the seed-level pattern without the dense seed-strip visualization.

Deployed-Selection Robustness in Forecasting

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Traffic-Hourly Top-k	0.50	7.016	[6.756, 7.283]	6.977	[6.706, 7.281]
Traffic-Hourly Top-k	1.00	24.651	[24.226, 25.079]	24.473	[24.001, 25.051]
Inventory	0.50	0.240	[0.195, 0.286]	0.202	[0.154, 0.256]
Inventory	1.00	1.029	[0.946, 1.111]	1.045	[0.926, 1.167]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A16. Paired-bootstrap intervals for deployed-gap rows. Deployed gap is the paired deployed-utility shortfall $U_{\text{best}} - U_{\text{forecast-selected}}$ computed within each matched seed. Intervals use 10,000 paired seed bootstrap resamples. Event-micro and Traffic-Hourly provide the main forecasting-native Q2 gap evidence; inventory is operational corroboration; load-following remains appendix-only directional context.

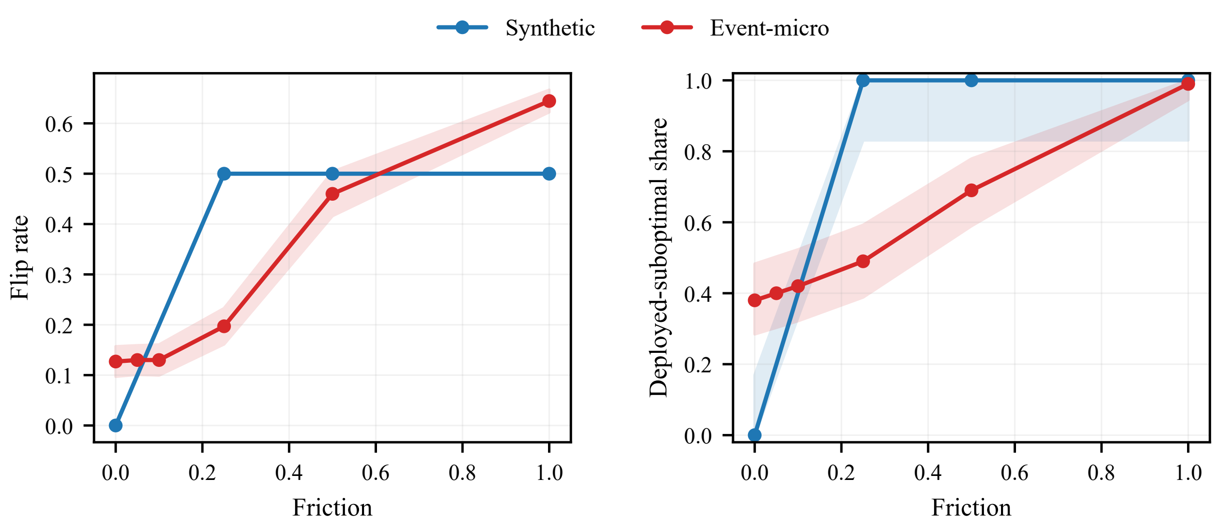


Figure A4. Q2 uncertainty summaries for synthetic and event-micro. Left: flip rate; right: deployed-suboptimal share. Bands follow the interval convention described in Appendix F; zero-width bands may coincide with the line.

F.1. Epsilon-Tie Sensitivity

The audit covers Event-micro, Traffic-Hourly Top-k, Inventory, and the appendix-only Load-following check; Load-following remains appendix-only because its lower-margin rows are tie-sensitive or partially sensitive.

Table A17. ϵ-tie sensitivity for the gated report-card domains. A seed is tie-compatible when the deployed shortfall of the forecast-selected model is at most $\epsilon_{\mathrm{rel}}S + 10^{-12}$, where S is the median seed-level deployed-utility scale within the row.

Domain	Interface	κ	ϵ_{rel}	n	S	Nom. subopt.	Adj. subopt.	Tie-comp.	Strict	Median Δ	Mean Δ	Verdict
Event-micro	threshold $\tau = 0.55$	0.50	0.000	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	0.50	0.001	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	0.50	0.005	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.000	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.001	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.005	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Traffic Top-k	k=249	0.50	0.000	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	0.50	0.001	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	0.50	0.005	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top-k	k=249	1.00	0.000	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top-k	k=249	1.00	0.001	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top-k	k=249	1.00	0.005	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Inventory	replenishment	0.50	0.000	100	4.841	0.74	0.74	26	74	0.202	0.240	stable
Inventory	replenishment	0.50	0.001	100	4.841	0.74	0.73	27	73	0.202	0.240	stable
Inventory	replenishment	0.50	0.005	100	4.841	0.74	0.72	28	72	0.202	0.240	stable
Inventory	replenishment	1.00	0.000	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.001	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.005	100	7.141	0.99	0.99	1	99	1.045	1.029	stable

Table A18. ϵ-tie sensitivity for the appendix-only Load-following secondary check. These rows are reported separately because Load-following is not used as primary evidence and includes lower-margin tie-sensitive rows.

Domain	Interface	κ	ϵ_{rel}	n	S	Nom. subopt.	Adj. subopt.	Tie-comp.	Strict	Median Δ	Mean Δ	Verdict
Load-following	responsive	0.25	0.000	10	0.046	0.70	0.70	3	7	0.000	0.001	tie-sensitive
Load-following	responsive	0.25	0.001	10	0.046	0.70	0.70	3	7	0.000	0.001	tie-sensitive
Load-following	responsive	0.25	0.005	10	0.046	0.70	0.50	5	5	0.000	0.001	tie-sensitive
Load-following	responsive	0.50	0.000	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	0.50	0.001	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	0.50	0.005	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	1.00	0.000	10	0.081	0.80	0.80	2	8	0.002	0.002	partially sensitive
Load-following	responsive	1.00	0.001	10	0.081	0.80	0.80	2	8	0.002	0.002	partially sensitive
Load-following	responsive	1.00	0.005	10	0.081	0.80	0.70	3	7	0.002	0.002	partially sensitive